BCS-061	SOFTWARE TESTING	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To discuss software testing background.
CO2	To introduce software testing techniques.
CO3	To explain different types of testing to understand realistic problem.
CO4	To develop analyzing techniques through automation testing tool.
CO5	To create awareness about the process part as per as software testing is concern.

Detailed Syllabus

MODULE I

Introduction

Faults, Errors, and Failures, Basics of software testing, Testing objectives, Principles of testing, Requirements, behavior and correctness, Testing and debugging, Test metrics and measurements, Verification, Validation and Testing, Types of testing, Software Quality and Reliability, Software defect tracking.

White Box and Black Box Testing

White box testing, static testing, Structural testing: Unit/Code functional testing, Code coverage testing, Code complexity testing, Black Box testing, Requirements based testing, Boundary value analysis, Equivalence partitioning, state/graph based testing, Model based testing and model checking, Differences between white box and Black box testing.

MODULE II

Integration, System, and Acceptance Testing

Top down and Bottom up integration, Bi-directional integration, System integration, Scenario Testing, Defect Bash, Functional versus Non-functional testing, Design/Architecture verification, Deployment testing, Beta testing, Scalability testing, Reliability testing, Stress testing, Acceptance testing: Acceptance criteria, test cases selection and execution,

MODULE III

Test Selection & Minimization for Regression Testing

Regression testing, Regression test process, Initial Smoke or Sanity test, Selection of regression tests, Execution Trace, Dynamic Slicing, Test Minimization, Ad hoc ,Testing: Pair testing, Exploratory testing, Iterative testing, Defect seeding.

Test Management and Automation

Test Planning, Management, Execution and Reporting, Software Test Automation: Scope of automation, Design & Architecture for automation, Generic requirements for test tool framework.

Text Books:

1. S. Desikan and G. Ramesh, "Software Testing: Principles and Practices", Pearson Education.
2. Aditya P. Mathur, "Fundamentals of Software Testing", Pearson Education.
3. Naik and Tripathy, "Software Testing and Quality Assurance", Wiley
4. K. K. Aggarwal and Yogesh Singh, "Software Engineering", New Age International Publication.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Formulate problem by following Software Testing Life Cycle.
CO2	Design Manual Test cases for Software Project.
CO3	Identify the realistic problem for different category of software.
CO4	Use automation testing tool students will be able test the software.
CO5	Follow the process related activity and testing techniques to work as team member